

Myung Chul (Michael) Kim

PhD Candidate, Mechanical Engineering | Stanford University | U.S. Permanent Resident

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Summary

Stanford Mechanical Engineering PhD candidate developing simulation and machine learning methods for strain and defects, from continuum finite element models to atomistic rare-event kinetics. Interested in advancing **semiconductor processing**, **packaging**, and **reliability** through physics-based modeling and accelerated sampling.

Education

Stanford University, PhD in Mechanical Engineering [GPA: 3.93/4.0] Sept. 2023 – June 2028

- Advisor: Prof. Wei Cai | **James D. Plummer Graduate Fellow** (2023–24)
- Coursework: Defects and Disorder in Materials, Finite Element Analysis, Elasticity and Inelasticity, Applied Quantum Mechanics, Numerical Linear Algebra, High Performance Computing

KAIST, BS in Mechanical Engineering and Electrical Engineering (double major) Mar. 2019 – Aug. 2023

Georgia Institute of Technology, Exchange Program in Mechanical Engineering Jan. 2022 – Dec. 2022

Research Experience

Doctoral Researcher, Stanford University – Stanford, CA Sept. 2023 – Present

Strain and stress modeling of Si–Ge–Sn core/multishell nanowires (with Stanford MSE and SLAC)

- Built a 3-D anisotropic **finite element** model (FEniCSx, Gmsh; 650,988 DOFs) using collaborators' STEM-EDS maps to quantify how composition and geometry determine strain and stress in semiconductor (Ge/GeSn/SiGeSn) nanostructures
- Verified the solver (patch test, analytical core-shell cylinder, equilibrium residual 2×10^{-13}) and derived the $\langle 111 \rangle$ normal-shear **stiffness coupling** that explains the three-fold shear pattern in collaborators' **4D-STEM** maps
- Formulated a two-channel (Sn/Si) EDS composition-error model calibrated against 4D-STEM strain data, enabling FEM predictions under various boundary and loading conditions without additional experimental characterization
- Ran shell-averaged and EDS-driven models to separate compositional from crystallographic strain, and showed that the in-plane dilatation is uncoupled from shear in the $\langle 111 \rangle$ frame, making it the robust scalar for 4D-STEM comparison

Accelerated atomistic simulations (PhD thesis, advisor W. Cai)

- Developed a descriptor-free **importance sampling** method for Langevin dynamics that uses neural networks to discover **rare transition pathways** and estimate their **timescales**
- Trained a graph neural network committor for **vacancy migration in bcc Fe** ($d = 2055$), which governs self-diffusion: within 3% of brute force at 600–800 K (uncorrected: 35–44% low) and, reused at 500 K, within 2% of its Arrhenius fit
- Resolved 12 transition modes of **Pt adatom diffusion** on Pt(001), which governs thin-film growth, with no collective variable: diagonal-to-crucifix ratio 6.27 and an unanticipated long-jump channel's 9.3% share (brute force: 6.27, 8.9%)
- Built the **PyTorch** GPU code behind these results as sole developer (autograd EAM forces, Verlet lists, torch.compile, SLURM arrays on A100/L40S GPUs), training committors on self-generated paths, so no labeled data is needed

Burgers-vector identification in dark-field X-ray microscopy (co-first author, theory and simulation)

- Derived a Burgers circuit on center-of-mass maps from two goniometer scans, toward non-destructive Burgers-vector identification, validated in a collaborator's simulator on all 12 signed $a/2 \langle 110 \rangle$ vectors in diamond (rms error $\leq 0.002a$)

Dislocation mobility from internal-stress statistics (co-first author, theory and simulation)

- Derived a closed-form velocity law for dislocation links in fcc Cu from internal-stress statistics of discrete dislocation dynamics, toward crystal-plasticity flow rules: $R^2 = 0.955$ with four fitted parameters vs. 0.965 for a 14-parameter fit

Mechanics of fibrin fiber networks (in progress, collaborators' tensile tests)

- Built a coarse-grained **Python/LAMMPS** model of 3-D fibrin networks to test whether fiber plasticity explains residual deformation in blood clots, calibrated to collaborators' load-unload tests (peak stress within 6%, loop area within 2%)

Undergraduate Research Assistant, Georgia Institute of Technology – Atlanta, GA Jan. 2022 – Dec. 2022

- Designed and fabricated a stretchable **piezoelectric pressure sensor** for electronic skin (*Adv. Mater. Technol.* 2025, cover) and developed a closed-loop respirator fit controller (*Biomaterials* 2025: fit factor 10% above a commercial one)

Publications

88 citations and h -index 3 as of Sept. 2026; two journal covers. scholar.google.com/citations?user=lzw1uB4AAAAJ

- **Accelerated Langevin Dynamics Simulation via Neural Network-Driven Importance Sampling** | **M. Kim**, W. Cai | *chemRxiv*, Mar. 2026 (revision with the Pt and Fe results in preparation)
- **Accelerated Markov Chain Monte Carlo Simulation via Neural Network-Driven Importance Sampling** | **M. Kim**, W. Cai | *arXiv:2602.12294*, Jan. 2026

- **Smart Filtering Facepiece Respirator with Self-Adaptive Fit and Wireless Humidity Monitoring** | K. Kwon, ..., M. C. Kim, ..., W.-H. Yeo | *Biomaterials*, 2025
- **Stretchable Wearable Wireless Bioelectronics Using All Printed Pressure Sensors and Strain Gauges** | N. Zavanelli, Y. J. Lee, M. C. Kim, ..., W.-H. Yeo | *Adv. Mater. Technol.*, 2025 | Journal cover
- **Advances in Electrochemical Sensors for Detecting Analytes in Biofluids** | J. Lee, M. C. Kim, I. Soltis, S. H. Lee, W.-H. Yeo | *Adv. Sensor Res.*, 2023 | Journal cover

Manuscripts in preparation (*equal contribution)

- **Shell-Resolved Elastic Strain Mapping in Radial Group-IV Multishell Nanowires** | A. Attiaoui, M. Kim, Á. Friðriksdóttir, W. Cai, C. Ophus, P. McIntyre
- **Burgers Vector Identification from Center-of-Mass Maps in Dark-Field X-Ray Microscopy** | M. Kim^{*}, B. Kanesalingam^{*}, Y. Wang, W. Cai, L. Dresselhaus-Marais
- **Dislocation Mobility from Internal Stress Fluctuations** (working title) | M. Kim^{*}, H. Zhai^{*}, S. Mohanty, W. Cai
- **Error-Correcting Importance Sampling for Descriptor-Free Acceleration of Rare Events: Application to Proteins** | M. Kim, S.-H. Ahn, W. Cai

Patent: A Method and Apparatus for Classification of Subtypes of Cells with Morphological and Motility Features using Hybrid Learning | H. Shin, ..., M. C. Kim | KR Patent 10-2762542, Jan. 2025

Presentations

ChemAI NYC (June 2026) | ICME Research Symposium (May 2026) | Stanford HAI (May 2026) | MRS Spring Meeting (Apr. 2026) | APS DAMOP (June 2025) | BMES Annual Meeting (Oct. 2022) | Gordon Research Conference (Sept. 2022)

Teaching and Mentoring

- **Teaching Assistant, ME 123 Computational Engineering** (Spring 2026; 40 students): ported core FEniCS examples to FEniCSx Colab notebooks and taught a Gmsh tutorial, to take students from finite differences to 3-D FEM in a browser; led sections, prepared exams, advised project teams; rated **4.61/5** overall ($n = 18$ of 40)
- **Teaching Assistant, ME 346A Introduction to Statistical Mechanics** (Winter 2025; 14 students; sole TA): led 8 problem sessions (two as exam reviews); wrote session notes and built Colab templates and solutions for Ising and virial-coefficient assignments, to help students connect theory to computation; rated **4.88/5** overall ($n = 8$ of 14)
- **Research Mentor** (2024–25; two visiting undergraduates): for a Stanford SURF 2025 student from UNAM (8 weeks) testing whether an ML potential transfers to large-scale MD, built the starter LAMMPS-to-ACE workflow, reviewed his code, and coached his symposium talk; onboarded a Univ. of Buenos Aires student (Winter 2024) to atomistic simulation

Awards and Honors

National Science & Technology Scholarship, Korea Ministry of Science and ICT (2019–23) | **Young Engineers Honors Society**, National Academy of Engineering of Korea (2020–23) | **Global Leadership Award**, KAIST (2023) | **President's Undergraduate Research Award**, Georgia Tech (2022) | **Student Design Finalist**, BMES and Medtronic (2022)

Technical Skills

Programming: Python, C, MATLAB, Bash, Git; PyTorch (autograd, `torch.compile`), NumPy, SciPy, Matplotlib

Machine Learning: SchNet-style graph neural networks; label-free training, replay buffers, target networks

Simulation Methods: importance sampling and variance reduction, Langevin and MD integrators (BAOAB, constrained dynamics), string method, harmonic TST, EAM and AMBER potentials, anisotropic FEM, FFT elasticity

Software and HPC: LAMMPS, OpenMM, FEniCSx, Gmsh, PETSc/MUMPS, ExaDiS; SLURM on A100/L40S GPUs | **Design:** SolidWorks, AutoCAD, Altium, PSpice | **Languages:** English (fluent), Korean (native)